
Plate-Local River Generation: Deterministic, Terrain-Aware, Downhill-Flowing Rivers for Infinite Procedural Worlds

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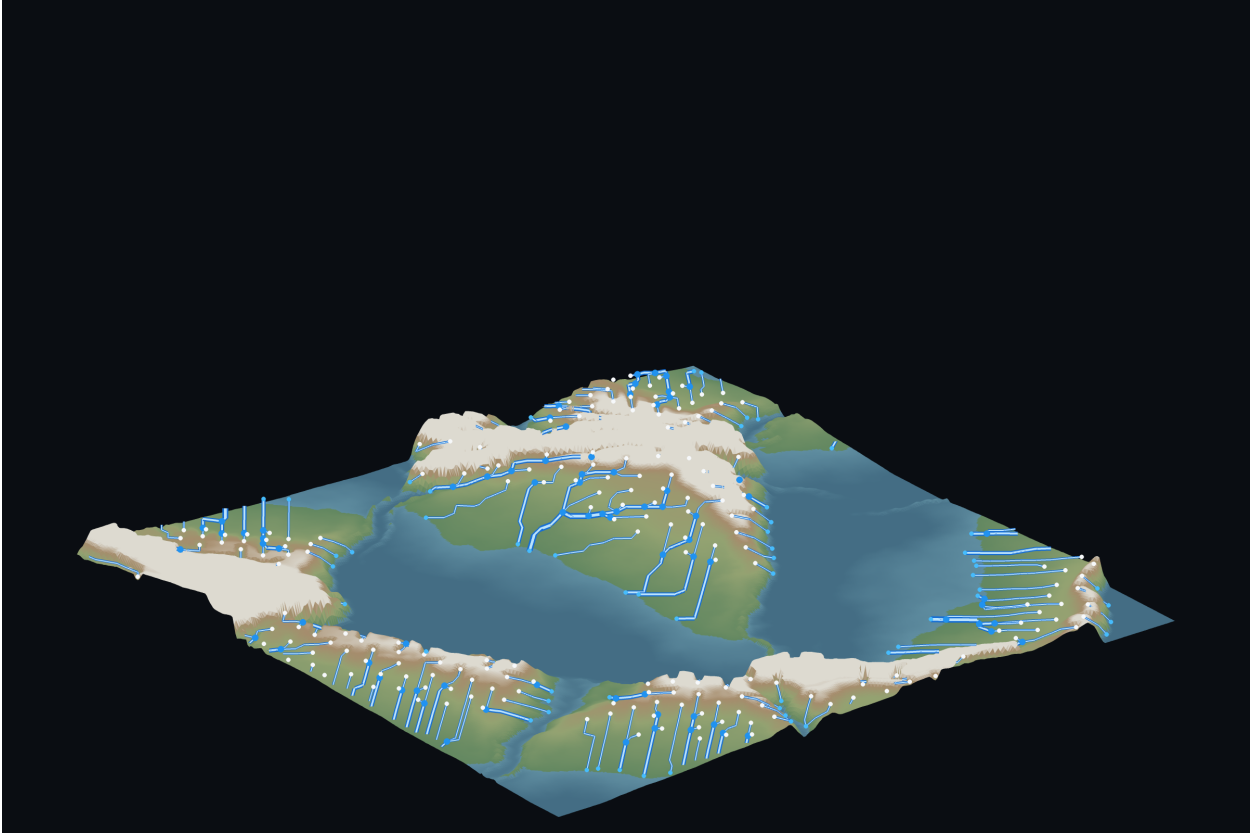


Figure 1: View sampled from the flat infinite-world implementation.

Existing procedural terrain generation methods produce coherent, downhill-flowing river networks only on fixed domains, whereas approaches that extend to infinite worlds typically sacrifice network coherence or downhill-flow guarantees. This work presents a deterministic, terrain-aware, downhill-flowing river algorithm for infinite procedural worlds. Tectonic plate seeds define both continuous terrain fields and corresponding bounded Voronoi-like plate domains, so localized river preprocessing can operate directly on the geometric domain without sampling the continuous terrain field to locate plate boundaries. Within each

plate, the terrain field is sampled on a low-resolution grid, and rivers are traced downhill from highland source nodes to coastlines, local minima, or existing confluences. The resulting river network, together with a geometric partition of the plate into nearest-river regions, is generated and cached lazily per plate, so each plate is processed independently of its neighbors. During full-resolution terrain sampling, a query point's nearest river segment is located directly from its cached region, and river height is interpolated along that segment and blended with the locally evaluated terrain on demand—without caching the blended

result itself—guaranteeing downhill flow. Together, these design choices make terrain-aware river generation practical for interactive procedural worlds by supporting efficient on-demand evaluation, as demonstrated by the performance analysis presented in this work.

1 Introduction

Procedural, deterministic, locally generatable terrain is well suited to large worlds because the terrain can be generated in independent regions that remain seamless with neighboring region. However, large-scale features such as river networks may depend on terrain relationships that extend far beyond the region currently being generated.

This work resolves that tension by using tectonic plates as an intermediate preprocessing domain: large enough to contain a plate’s regional drainage structure, but small enough to generate and cache independently of the rest of the world. Because plate boundaries are defined identically for terrain evaluation and river preprocessing, a plate’s river network can be built directly from its geometric domain, without first sampling or rasterizing the surrounding terrain. Because rivers are generated independently for each plate, they do not cross plate boundaries. This limitation is mitigated by constructing the continuous terrain fields so that oceans, plateaus, and mountain ranges naturally discourage cross-boundary drainage.

The key contributions of this work are:

- dual plate representation for terrain sampling and river preprocessing, described in Sections 3.2–3.4;
- lazy local river generation and caching, described in Section 3.7;
- nearest-river regions that accelerate river-segment lookup during chunk generation, described in Section 3.7.

2 Related Work

Related works for fixed domains generally fall into two broad categories: systems that construct rivers first, then the terrain is constructed around them; and systems that construct the terrain first, then the rivers

are constructed along this terrain. Learned terrain generation offers a novel approach to world generation in general, and can be extended to infinite domains [13].

2.1 River-First Generation.

River-first approaches generate river network structures before the surrounding terrain [10–12]. By treating rivers as the primary organizing structure, these methods guarantee coherent, downhill-flowing rivers while allowing explicit control over river placement, connectivity, and slope. However, because terrain is constructed around a predefined river network, large-scale geological processes such as tectonic uplift and mountain building are secondary to river constraints. Furthermore, these approaches require preprocessing over a fixed domain and are not readily extendable to infinite procedural worlds.

2.2 Terrain-First Generation.

Terrain-first approaches first generate terrain before deriving river networks from the resulting topology. Batbold and Ziegelmeier [9] demonstrate that coherent river networks can be derived from procedurally generated terrain using graph-based flow routing. The present work builds on this terrain-first philosophy with river generation derived from a low-resolution sample of continuous terrain functions, which is localized to plate domains. This enables river generation without preprocessing the entire domain.

2.3 Learned Generation.

InfiniteDiffusion [13] resolves the infinite-domain limitation of both river-first and terrain-first methods. Current limitations of diffusion-based terrain generators include limited explicit control over large-scale terrain features, such as continent placement, mountain range orientation, or coastline shape and no guarantee of coherent, downhill-flowing river networks. These limitations are not necessarily fundamental to diffusion models, but do remain unaddressed.

3 Plate-Local River Generation

The following subsections define deterministic plate seeds and their dual pointwise and geometric representations, then describe plate-local terrain sampling,

river tracing, caching, and on-demand composition. Together, these stages preserve terrain-aware downhill drainage without a global rasterization or preprocessing pass. For applications to finite, closed-shaped worlds such as spherical planet domains, we recommend slight modifications to this system, which we provide in this section alongside the infinite application.

Algorithms 1 and 2 detail the complete procedures for building plate-local river caches and evaluating terrain on demand.

Algorithm 1. *Plate-Local River Cache Build*

Require: Plate i , seed positions $\{\mathbf{c}_j\}$, surface $\mathcal{S}$
Ensure: River network $\mathcal{R}_i$ and nearest-river regions
Geometric plate domain (Section 3.4)
1: Initialize bounding volume B_i around $\mathbf{c}_i$
2: **for** each neighboring plate $j \in \mathcal{N}_i$ **do**
3: $\mathbf{m}_{ij} \leftarrow \frac{1}{2}(\mathbf{c}_i + \mathbf{c}_j)$; $\mathbf{n}_{ij} \leftarrow \mathbf{c}_j - \mathbf{c}_i$
4: Clip B_i by $(\mathbf{p} - \mathbf{m}_{ij}) \cdot \mathbf{n}_{ij} \leq 0$
5: **end for**
6: $\Omega_i^{\mathcal{S}} \leftarrow B_i \cap \mathcal{S}$
Low-resolution plate grid (Section 3.5)
7: **for** each (u, v) with $\mathbf{x}_{uv} \in \Omega_i^{\mathcal{S}}$ **do**
8: $h_{uv} \leftarrow A_{\text{pre}} H_{\text{pre}}(\mathbf{x}_{uv})$
9: **end for**
River node and segment generation (Section 3.6)
10: $\mathcal{N} \leftarrow \emptyset$; $\mathcal{R}_i \leftarrow \emptyset$
11: Select eligible highland sources outside the border mask
12: Add the selected source nodes to $\mathcal{N}$
13: **for** each source node $n \in \mathcal{N}$ **do**
14: Trace downhill from n along grid gradients
15: Stop at sea level, a local minimum, or an existing segment
16: Add the resulting segments to $\mathcal{R}_i$
17: **end for**
18: Assign Strahler orders after establishing all confluences
Nearest-river partitioning (Section 3.7)
19: Partition $\Omega_i^{\mathcal{S}}$ into $\{V_s\}_{s \in \mathcal{R}_i}$
20: Cache the network, regions, elevations, and stream orders

Algorithm 2. *On-Demand Terrain Sample*

Require: Position $\mathbf{x}$, seed positions $\{\mathbf{c}_j\}$, river cache
Ensure: Final height $H_{\text{final}}(\mathbf{x})$
Plate assignment (Section 3.3)

1: $\tilde{\mathbf{x}} \leftarrow s_p \mathbf{x} + \mathbf{w}(\mathbf{x})$
2: $i \leftarrow \arg \min_{j \in \mathcal{A}} \|\tilde{\mathbf{x}} - \mathbf{c}_j\|^2$ ▷ Owning plate
Cache lookup and lazy build (Section 3.7)
3: **if** the cache for plate i is absent **then**
4: Execute Algorithm 1 for plate i
5: **end if**
Pointwise terrain fields (Section 3.3)
6: Evaluate W_{land} , W_{mountain} , W_{plateau} , and W_{island} at $\mathbf{x}$
7: Compute $H_{\text{pre}}(\mathbf{x})$ from the weighted fields
River query (Section 3.7)
8: **if** $\mathcal{R}_i = \emptyset$ **then**
9: **return** $H_{\text{pre}}(\mathbf{x})$
10: **end if**
11: Use the region partition to identify $s^*(\mathbf{x})$
12: $\delta_{\text{river}}(\mathbf{x}) \leftarrow \text{dist}(\mathbf{x}, s^*(\mathbf{x}))$
13: Interpolate $H_{\text{river}}(\mathbf{x})$ along $s^*(\mathbf{x})$
Terrain composition (Section 3.8)
14: $\lambda_{\text{river}}(\mathbf{x}) \leftarrow \left[\min\left(1, \frac{\delta_{\text{river}}(\mathbf{x})}{\omega_{\text{river}}}\right) \right]^{\alpha_{\text{river}}}$
15: $H_{\text{final}}(\mathbf{x}) \leftarrow \lambda_{\text{river}}(\mathbf{x}) H_{\text{pre}}(\mathbf{x}) + (1 - \lambda_{\text{river}}(\mathbf{x})) H_{\text{river}}(\mathbf{x})$
16: **return** $H_{\text{final}}(\mathbf{x})$

3.1 Deterministic Tectonic Plate Positions

Each tectonic plate is defined by a deterministic seed position. The same construction can be used for finite and infinite worlds: finite worlds use a bounded set of plate identifiers, while infinite worlds generate plate identifiers from lattice cells on demand. In both cases, terrain samples are assigned to the nearest deterministic seed position, producing Voronoi-like plate regions [1, 2].

For an infinite world, plate seeds may be generated as one deterministic point per integer lattice cell, following the same nearest-feature-point structure used by cellular texture bases [6],

$$\mathbf{c}_i = s_p (\mathbf{i} + \boldsymbol{\eta}(\mathbf{i})), \quad \mathbf{i} \in \mathbb{Z}^d, \quad (1)$$

where s_p is the plate scale, i is the plate identifier associated with integer lattice-cell coordinate $\mathbf{i}$, and $\boldsymbol{\eta}(\mathbf{i})$ is a deterministic hash offset. For a finite world, with a closed geometry and N plates, a golden-angle sequence gives approximately even seed directions [3, 4]:

$$\theta_i = i\pi(3 - \sqrt{5}), \quad y_i = 1 - \frac{2i+1}{N}, \quad r_i = \sqrt{1 - y_i^2}, \quad (2)$$

$$\mathbf{q}_i = (r_i \cos \theta_i, y_i, r_i \sin \theta_i), \quad (3)$$

$$\mathbf{c}_i = s_p \Pi_{\mathcal{S}}(\mathbf{q}_i) + \boldsymbol{\eta}_i, \quad 0 \leq i < N, \quad (4)$$

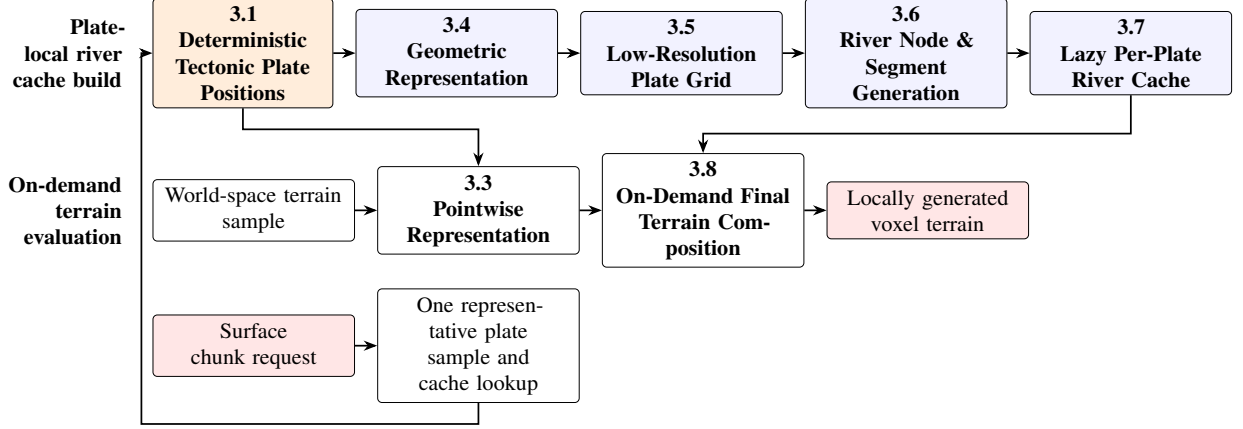


Figure 2: Pipeline overview, generalized for various world types—infinite or finite, and landscape structures—voxels or mesh.

Here $\Pi_{\mathcal{S}}$ projects the unit direction onto the chosen finite world surface, and η_i is a deterministic jitter applied to the finite plate center.

3.2 Dual Tectonic Plate Representation

The tectonic seed positions are used in two forms: a *pointwise representation*, defined by continuous coordinate-based fields for terrain evaluation, and a *geometric representation*, defined by a bounded non-rasterized domain for regional river preprocessing. The same seed positions drive both local terrain fields and plate domains for drainage-network generation, allowing individual plates to be processed without rasterizing the surrounding world.

3.3 Pointwise Representation

During chunk generation, terrain is evaluated directly from world-space coordinates by comparing plate seed positions and deriving continuous scalar and vector weight fields from plate ownership, plate type, border distance, and drift direction. These weight fields control land, mountain, plateau, and island chain placement. Noise warping and surface roughness are applied to produce more organic continent shapes [5].

Pointwise Plate. For a world-space point $\mathbf{x}$, the pointwise plate fields are evaluated in warped plate space,

$$\tilde{\mathbf{x}} = s_p \mathbf{x} + \mathbf{w}(\mathbf{x}), \quad (5)$$

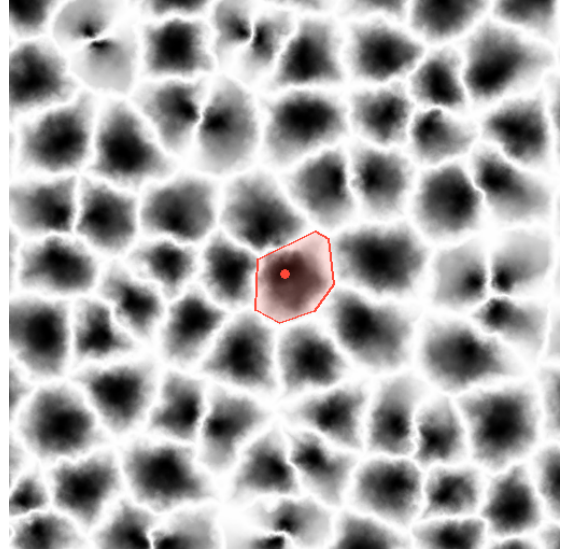


Figure 3: Dual plate representations. A continuous terrain field (grayscale) and a corresponding geometric plate domain (red) are independently derived from the same tectonic plate seed.

where $\mathbf{w}$ is a deterministic noise warp. Note that $\mathbf{w}$ cannot be represented in the geometric representation of the same plate. Given deterministic plate centers $\mathbf{c}_i$, the owning plate is the nearest center in this space:

$$i = \arg \min_{j \in \mathcal{A}} \|\tilde{\mathbf{x}} - \mathbf{c}_j\|^2. \quad (6)$$

Here $\mathcal{A}$ is the set of possible plate identifiers. For a finite world, $\mathcal{A} = \{1, \dots, N\}$ and each identifier names one generated plate seed. For an infinite world, $\mathcal{A} = \mathbb{Z}^d$ and each identifier is an integer lattice-cell

coordinate that generates one seed on demand. Plate fields are controlled by the distance margin between the owning plate and a neighboring plate:

$$d_j(\mathbf{x}) = \|\tilde{\mathbf{x}} - \mathbf{c}_j\| - \|\tilde{\mathbf{x}} - \mathbf{c}_i\|, \quad j \neq i. \quad (7)$$

Small values of d_j indicate proximity to the boundary between the owning plate and plate j . When a terrain feature depends on the nearest eligible neighboring plate, the margin is minimized over a selected neighbor set $\mathcal{N}$:

$$d_{\mathcal{N}}(\mathbf{x}) = \min_{j \in \mathcal{N}} d_j(\mathbf{x}). \quad (8)$$

Plate Drift. Each plate identifier is assigned a deterministic drift axis $\mathbf{a}_i$ by projecting a random vector into the tangent plane of the terrain domain. For a non-flat world, such as a finite one with a closed geometry, the plate center direction vector from the world origin provides a stable reference normal:

$$\mathbf{a}_i = \frac{\mathbf{c}_i \times \mathbf{r}_i}{\|\mathbf{c}_i \times \mathbf{r}_i\|}. \quad (9)$$

For a flat world, such as an infinite one, the constant reference normal $\mathbf{n}_0$ can be used instead:

$$\mathbf{a}_i = \frac{\mathbf{n}_0 \times \mathbf{r}_i}{\|\mathbf{n}_0 \times \mathbf{r}_i\|}. \quad (10)$$

The drift direction at the terrain sample is then

$$\ell(\mathbf{x}) = \frac{\mathbf{a}_i \times \mathbf{n}(\mathbf{x})}{\|\mathbf{a}_i \times \mathbf{n}(\mathbf{x})\|}, \quad (11)$$

where $\mathbf{r}_i$ is the deterministic random vector assigned to plate identifier i , $\mathbf{n}(\mathbf{x})$ is the local surface normal, and $\ell(\mathbf{x})$ is the local tangent drift direction used at the terrain sample. The scalar drift field, $\rho(\mathbf{x})$, measures signed progress along the local drift direction:

$$\rho(\mathbf{x}) = (\tilde{\mathbf{x}} - \mathbf{c}_i) \cdot \ell(\mathbf{x}). \quad (12)$$

Weights. Land, island, and mountain feature weights use the same normalized beta-shaped border profile. For feature k with amplitude A_k , width ω_k , interior exponent p_k , border exponent q_k , and

plate-coverage parameter γ_k ,

$$C_k = \left[\left(\frac{p_k}{p_k + q_k} \right)^{p_k} \left(\frac{q_k}{p_k + q_k} \right)^{q_k} \right]^{-1}, \quad (13)$$

$$D_k(\mathbf{x}) = \max \left(0, \frac{\rho(\mathbf{x}) + 2\gamma_k - 1}{2\gamma_k} \right), \quad (14)$$

$$z_k(\mathbf{x}) = \frac{d_{\mathcal{N}_k}(\mathbf{x})}{\omega_k}, \quad (15)$$

$$W_k(\mathbf{x}) = \begin{cases} A_k C_k D_k(\mathbf{x}) z_k(\mathbf{x})^{p_k} \cdot (1 - z_k(\mathbf{x}))^{q_k}, & 0 \leq z_k(\mathbf{x}) \leq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

The coefficient C_k normalizes the profile before amplitude scaling, and D_k gates the feature by the drift field.

Plateaus are treated separately because they arise from converging continental plates rather than a single boundary-distance profile. For a nearby continental pair (i, j) , let $\hat{\mathbf{u}}_{ij}(\mathbf{x})$ denote the unit separation direction from plate i to plate j after projection into the local tangent plane. The closing response is measured by projecting each plate's tangent drift onto this direction:

$$\kappa_{i \rightarrow j}(\mathbf{x}) = \begin{cases} (\mathbf{a}_i \times \mathbf{n}(\mathbf{x})) \cdot \hat{\mathbf{u}}_{ij}(\mathbf{x}), & \text{if positive,} \\ 0, & \text{if zero or negative,} \end{cases} \quad (17)$$

$$\kappa_{j \rightarrow i}(\mathbf{x}) = \begin{cases} -(\mathbf{a}_j \times \mathbf{n}(\mathbf{x})) \cdot \hat{\mathbf{u}}_{ij}(\mathbf{x}), & \text{if positive,} \\ 0, & \text{if zero or negative.} \end{cases} \quad (18)$$

These detect colliding plates. The weight itself is constructed as,

$$R_{\text{plateau}}(\mathbf{x}) = O(\mathbf{x}) \max_{(i,j) \in \mathcal{C}} B_{ij}(\mathbf{x}) N_i(\mathbf{x}) \cdot \kappa_{i \rightarrow j}(\mathbf{x}) \kappa_{j \rightarrow i}(\mathbf{x}) \cdot S\left(\frac{\kappa_{i \rightarrow j}(\mathbf{x})}{\eta}\right) S\left(\frac{\kappa_{j \rightarrow i}(\mathbf{x})}{\eta}\right), \quad (19)$$

$$W_{\text{plateau}}(\mathbf{x}) = A_{\text{plateau}} S(g_{\text{plateau}} R_{\text{plateau}}(\mathbf{x})) \cdot (A_{\text{land}} - W_{\text{land}}(\mathbf{x})) \cdot (A_{\text{mountain}} - W_{\text{mountain}}(\mathbf{x})). \quad (20)$$

Here $\mathcal{C}$ is the set of nearby continental plate pairs, B_{ij} fades with the distance between the plate centers, N_i fades with the distance from the owning plate, O suppresses plateau formation near oceanic boundaries, η

is the closing fade width, g_{plateau} controls the plateau response gain, and S is a smoothstep function. The final plateau weight W_{plateau} is an amplitude-scaled smoothstep response to the closing response R_{plateau} , suppressed where the land and mountain fields already dominate.

3.4 Geometric Representation

The entire plate domain can be constructed geometrically from nearby seed positions without pointwise terrain sampling. Because the noise warping applied to the weight fields cannot be reproduced geometrically, this domain is a straight-bordered approximation of the true plate footprint—close enough for river generation purposes.

The geometric plate domain is initialized as a bounded volume B_i around the plate center and then clipped by the bisector planes between $\mathbf{c}_i$ and $\mathbf{c}_j$:

$$\Omega_i = B_i \cap \bigcap_{j \neq i} \{ \mathbf{p} : (\mathbf{p} - \mathbf{m}_{ij}) \cdot \mathbf{n}_{ij} \leq 0 \}, \quad (21)$$

where

$$\mathbf{m}_{ij} = \frac{1}{2} (\mathbf{c}_i + \mathbf{c}_j), \quad (22)$$

$$\mathbf{n}_{ij} = \mathbf{c}_j - \mathbf{c}_i. \quad (23)$$

Note that $j \in \mathcal{N}_i$, where $\mathcal{N}_i$ is finite. For a world surface $\mathcal{S}$, the plate-local preprocessing domain is the surface cross section of this clipped volume,

$$\Omega_i^{\mathcal{S}} = \Omega_i \cap \mathcal{S}. \quad (24)$$

The Sutherland-Hodgman polygon clipping algorithm is used for this step in the implementation [7]. Recall that this geometric representation is not identical to the pointwise representation due to the warp term $\mathbf{w}$ in Equation (5).

3.5 Low-Resolution Plate Grid

The river generator samples pre-river terrain using the weight fields on a low-resolution grid restricted to the geometric representation domain, providing approximate terrain elevation and slope for river path planning. Using a low-resolution terrain sample makes plate-local river preprocessing computationally practical by significantly reducing computation time and

memory requirements, while still retaining a terrain-first generation philosophy.

For each accepted grid sample (u, v) on plate i , the sample position lies in the geometric surface domain and stores the pre-river terrain height:

$$\begin{aligned} \mathbf{x}_{uv} &\in \Omega_i^{\mathcal{S}}, \\ h_{uv} &= A_{\text{pre}} H_{\text{pre}}(\mathbf{x}_{uv}), \end{aligned} \quad (25)$$

where $A_{\text{pre}} < 1$ is a pre-river height attenuation factor. By reducing the terrain height used for river planning, the subsequently enforced river elevations are biased below the surrounding terrain surface, producing valley-like depressions after composition. A_{pre} may be defined to attenuate land, mountain, plateau, and island features differently.

3.6 River Node & Segment Generation

River source nodes are selected from grid points of sufficient elevation. Segments are traced downhill along the grid until reaching sea level, a local minimum, or an existing river segment, producing outlets, lakes, or confluences respectively.

Note that the algorithm for placing nodes and segments is arbitrary and can be modified or replaced without affecting the rest of the pipeline. However, to avoid blending discontinuities during final terrain composition, nodes and segments should not be placed near plate borders. The river blending width, ω_{river} (see Section 3.8), defines a proximity mask for removing or preventing node placement near these borders. Because river-terrain blending does not extend beyond ω_{river} from a given river, this constraint prevents river blending from crossing plate borders and introducing discontinuities.

3.7 Lazy Per-Plate River Cache

A plate’s river network is generated the first time a chunk samples terrain from that plate and finds the cache absent. The cache stores river nodes, segments, and elevations, along with a non-rasterized geometric partitioning of the plate domain into nearest-river regions. These nearest-river regions allow chunk generation to locate the nearest river segment efficiently without searching the full network. Because rivers naturally should not cross tectonic boundaries—where

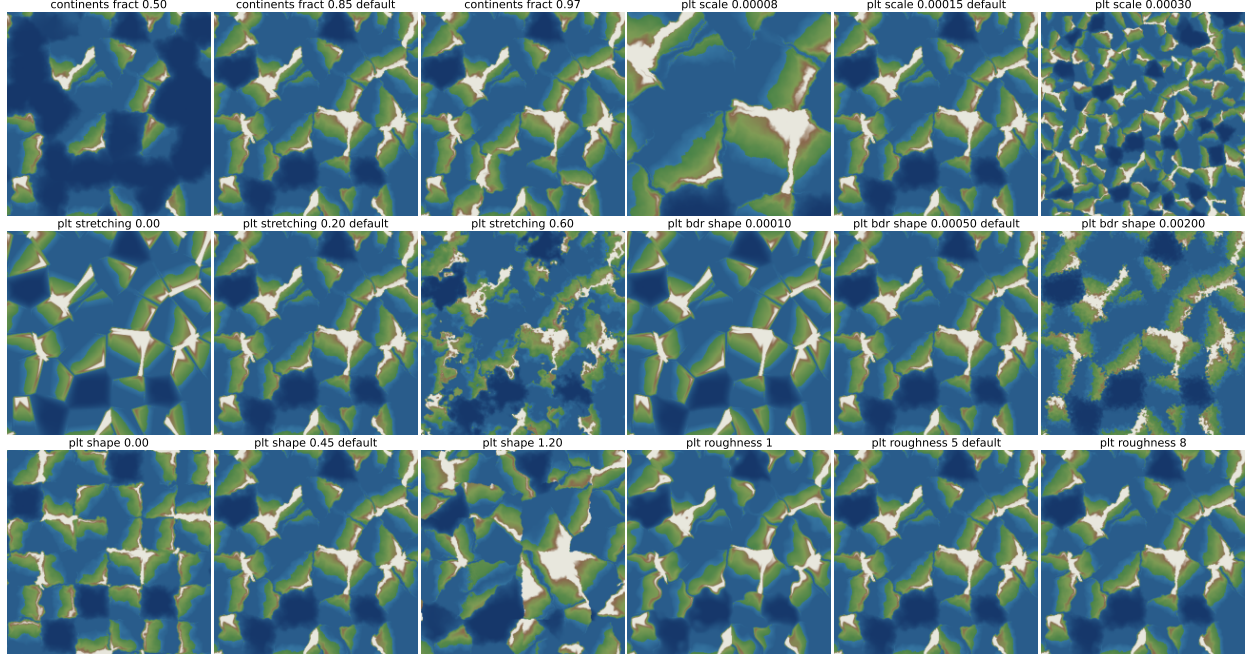


Figure 4: Top-down height-field renders for one fixed seed, sweeping one plate-tessellation parameter per row (low, default, high) with all other parameters held at their Table-listed defaults. Row 1: oceanic/continental threshold, then plate scale s_p . Row 2: warp amplitude, then warp spatial frequency. Row 3: plate-seed jitter (zero jitter collapses to a regular lattice), then warp octave count.

plateaus and oceans generate within this system—each plate’s network can be generated and cached in isolation without triggering generation of neighboring plates.

For a terrain sample on plate i , let $\mathcal{R}_i$ be the cached set of river segments for that plate. The nearest segment and its distance are

$$\begin{aligned} s^*(\mathbf{x}) &= \arg \min_{s \in \mathcal{R}_i} \text{dist}(\mathbf{x}, s), \\ \delta_{\text{river}}(\mathbf{x}) &= \text{dist}(\mathbf{x}, s^*(\mathbf{x})). \end{aligned} \quad (26)$$

Additional river fields can be evaluated from the same nearest segment. In particular, the Strahler stream order [8] provides a deterministic river-size field by assigning source tributaries order one and increasing the order at confluences where equal-order tributaries meet. For a sample $\mathbf{x}$, the size field may therefore be taken from the nearest segment,

$$\sigma_{\text{river}}(\mathbf{x}) = \text{strahler}(s^*(\mathbf{x})). \quad (27)$$

Small deterministic noise offsets may also be applied to the river distance, height, or size fields before final

composition. When the same seed and coordinate basis are used for all chunks, these perturbations make river channels appear more curved while preserving deterministic local generation.

3.8 On-Demand Final Terrain Composition

During chunk generation, plate weight fields are sampled, distance to the nearest river segment is computed, and river elevation is interpolated between nodes. The river distance is converted into a normalized terrain blend weight before mixing the local terrain height with the river height:

$$\begin{aligned} \lambda_{\text{river}}(\mathbf{x}) &= \left[\min \left(1, \frac{\delta_{\text{river}}(\mathbf{x})}{\omega_{\text{river}}} \right) \right]^{\alpha_{\text{river}}}, \\ H_{\text{final}}(\mathbf{x}) &= \lambda_{\text{river}}(\mathbf{x}) H_{\text{pre}}(\mathbf{x}) + (1 - \lambda_{\text{river}}(\mathbf{x})) H_{\text{river}}(\mathbf{x}). \end{aligned} \quad (28)$$

Here ω_{river} is the river blending width and α_{river} controls the blend falloff. Near the river, λ_{river} approaches zero and the interpolated river height dominates; away from the river, λ_{river} approaches one and the locally generated terrain is preserved.

4 Experimental Results and Evaluation

A video demonstration for a finite, cube-shaped planet implementation of this system is available on YouTube at youtube.com/watch?v=4Ai_13znvgg.

Code for a toy example similar to Figure 1 is available at github.com/Davismk3/plate-local-river-generation.

Figure 1 and Table 1 show output and measured performance for a single deterministic run, detailed below. All benchmark code and raw results accompany this submission; every statistic is computed directly from the released `flat-infinite` implementation.

The implementation is written in Python with OpenGL and PyGame rendering. No game engine is used. Hot functions use Numba `@njit` [14]; additional optimizations are used elsewhere.

4.1 Reference Configuration

The figure exporter also returns a JSON summary and a CSV file from the same run. All reported measurements correspond to Figure 1. The figure is not generated separately. The run uses seed 1 and a square view two plate-cell widths across. The terrain is sampled at a resolution of 320×320 (102,400 samples). Eleven plate caches intersect the view. River networks are generated on a 100×100 grid within each plate. At most 50 river paths are requested per plate. Channel lengths are normalized to plate-cell widths.

The sampled terrain contains 60.17% land and 39.83% ocean at a sea level of 0.35. Terrain heights range from 0.200 to 1.445. The median height is 0.428. The 95th percentile height is 1.200.

The 11 plate caches contain 2,063 nodes and 1,853 segments. Of the 550 requested river paths, 440 are accepted (80.0%). The networks contain 426 source nodes, 208 sea outlets, two local-minimum outlets, 197 confluences, and 210 connected components.

The total channel length across all plate caches is 64.888 plate-cell widths. The average channel length per plate is 5.899 plate-cell widths. Individual plates contain 115–222 segments and 28–50 accepted river paths. Their total channel lengths range from 4.088

to 7.771 plate-cell widths.

Every stored river segment was tested using its adjusted endpoint river heights. No uphill segments were found. Downhill compliance is 100

The nearest-segment partition contains 1,853 regions, one for each river segment. These regions are represented by 10,743 polygon vertices in total.

After clipping the river network to the meshed domain shown in Figure 1, 945 nodes and 812 segments remain. The visible network contains 231 source nodes, 89 sea outlets, two local-minimum outlets, and 623 channel nodes. The visible channel length is 28.361 plate-cell widths.

The packed numeric cache for the 11 plate caches occupies 0.502 MiB. This corresponds to approximately 284 bytes per river segment. Python object overhead used during preprocessing is excluded.

The measurements were collected on an Apple M4 Max system. Construction of the 11 plate caches requires 4.376s in total. The average construction time is 0.398s per cache. The median is 0.408s. Individual cache construction times range from 0.184 to 0.621s. Cache packing requires 0.008s. Terrain sampling and visible-geometry generation require 0.315s. Rendering and PNG export require 1.689s. Rendering includes $3 \times$ supersampling for anti-aliasing, down-sampling to the final image resolution, and rasterization of an auxiliary depth buffer for terrain occlusion. The total export time is 6.631s. These timings are observational rather than controlled benchmarks. They may vary with compilation state and system load. The terrain and river network statistics are deterministic.

Table 1. Metrics emitted by the same deterministic exporter invocation that produced Figure 1. Lengths are measured in plate-cell widths. Complete-network counts include all 11 plate caches intersecting the view; visible counts are clipped to the displayed mesh.

Metric	Value
Terrain samples	102,400
Visible plate caches	11
Land / ocean fraction	60.17 / 39.83%
Median / 95th-percentile height	0.428 / 1.200

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Table 1 continued

Metric	Value
Nodes / segments	2,063 / 1,853
Accepted / requested paths	440 / 550
Sources / confluences	426 / 197
Sea / local-minimum outlets	208 / 2
Connected components	210
Downhill violations	0
Total / visible channel length	64.888 / 28.361
Visible nodes / segments	945 / 812
Regions / polygon vertices	1,853 / 10,743
Packed-cache size	0.502 MiB
Cache build total / mean	4.376 / 0.398 s
Pack / sample / render	0.008 / 0.315 / 1.689 s

4.2 Future Work

We recommend that future implementations of this system be written in a compiled language like C++, not Python. Python was selected for its familiarity at the time.

The plate-local framework is not limited to rivers. The same bounded-domain pattern could generate roads, towns, or other features that guaranteed a certain topology. Each application would replace downhill flow with its own constraints—for example slope and travel cost for roads, or water access and terrain suitability for towns—while retaining deterministic identifiers, lazy regional preprocessing, packed caches, and pointwise evaluation.

Most notably, integration with learned generators such as InfiniteDiffusion [13] offers a longer-term hybrid direction. Plate-local fields and river graphs could supply explicit structural constraints, while a learned model adds fine surface, coastline, and mountain detail.

4.3 Conclusion

We have presented an algorithm for lazily generating river networks that are terrain-aware, deterministic, and guarantee coherence and downhill flow, which can be tailored to flat infinite world-types or closed-shaped finite world-types such as spheres and cubes.

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